IILM UNIVERSITY, GREATER NOIDA
SCHOOL OF COMPUTER SCIENCE & ENGINEERING
WEEKLY PROGRESS REPORT
Even semester 2026-2027

B.Tech. Major Project Details

Project Id	**26P_3CSE389**
Project Title	**Reinforcement Learning-Based Dynamic Battery Thermal Management for Software Defined Electric Vehicles**
Reporting period	_____
Week No.	**1**

Work Completed This Week:

- Reviewed the battery thermal-management problem and the trade-off between temperature safety, spatial uniformity and auxiliary cooling energy.
- Reviewed reinforcement-learning thermal-control literature and conventional PID and model-predictive control approaches for the project scope.
- Specified a simulation-based design using a four-node battery pack, an electrical equivalent-circuit model and a finite coolant inventory.
- Identified pump, fan and chiller as three independent continuous control inputs; selected PPO, SAC and TD3 for comparison.
- Identified official EPA-hosted UDDS, HWFET, US06, ECE and EUDC speed traces and the ECE/EUDC assembly required for NEDC.
- Prepared the implementation and evaluation workflow, covering model checks, fair baseline tuning, multi-seed training and separate raw/supervised evaluation.

Challenges/Issues:

- Selecting a useful model fidelity while clearly distinguishing engineering assumptions from measured battery parameters.
- Defining a common objective and evaluation protocol so controller comparisons remain fair and reproducible.
- Keeping simulation, computational feasibility and physical validation claims distinct.

Plan for Next Week: Implement the coupled vehicle, battery and cooling models; prepare real driving inputs and verify limiting-case behaviour before controller training.

Guide's Remarks:

Signature of Student
Harshit Singh Parmar

Signature of Guide
[Dr. Pranav Srivastava]

Retrospective milestone grouping; reporting dates and weekly allocation require confirmation.

IILM UNIVERSITY, GREATER NOIDA
SCHOOL OF COMPUTER SCIENCE & ENGINEERING
WEEKLY PROGRESS REPORT
Even semester 2026-2027

B.Tech. Major Project Details

Project Id	**26P_3CSE389**
Project Title	**Reinforcement Learning-Based Dynamic Battery Thermal Management for Software Defined Electric Vehicles**
Reporting period	_____
Week No.	**2**

Work Completed This Week:

- Set up the local CPU Python workflow using NumPy, Gymnasium, PyTorch, Stable-Baselines3 and plotting utilities.
- Implemented the longitudinal vehicle-demand model and first-order battery equivalent circuit, including state of charge, polarization dynamics and electrical heat generation.
- Developed four thermal nodes with neighbour conduction, ambient exchange and serial coolant warming; coupled them to a finite coolant reservoir.
- Implemented separate pump, radiator-fan and chiller commands, including their electrical energy requirements and pump-off circulation behaviour.
- Loaded the retained official driving traces, assembled NEDC from ECE and EUDC, and preserved source units and file hashes for provenance.
- Checked heat balance, cooling response, charging/discharging signs, equilibrium and normalizer behaviour using the plant test cases.

Challenges/Issues:

- Converting speed traces into modeled battery power/current while accounting for drivetrain efficiency, accessories and bounded regeneration.
- Maintaining consistent electrical and thermal energy accounting, particularly polarization-capacitor storage and finite coolant capacity.
- Documenting thermal and cooling assumptions without presenting them as physical-pack measurements.

Plan for Next Week: Build the Gymnasium control environment, implement the shared cost and safety supervisor, and establish conventional baseline performance.

Guide's Remarks:

Signature of Student
Harshit Singh Parmar

Signature of Guide
[Dr. Pranav Srivastava]

Retrospective milestone grouping; reporting dates and weekly allocation require confirmation.

IILM UNIVERSITY, GREATER NOIDA
SCHOOL OF COMPUTER SCIENCE & ENGINEERING
WEEKLY PROGRESS REPORT
Even semester 2026-2027

B.Tech. Major Project Details

Project Id	**26P_3CSE389**
Project Title	**Reinforcement Learning-Based Dynamic Battery Thermal Management for Software Defined Electric Vehicles**
Reporting period	_____
Week No.	**3**

Work Completed This Week:

- Built the custom Gymnasium environment with reset, step and termination logic, using 1-second plant updates and 5-second controller decisions.
- Defined twelve observation values, including four module temperatures, state of charge, C-rate, ambient/coolant conditions, demand and preview summaries.
- Implemented three bounded actuator commands and a common cost covering temperature-band exceedance, spatial spread, cooling electricity and relative aging exposure.
- Added a heuristic full-cooling supervisor at 42 C and kept raw controller requests distinct from the commands actually applied.
- Implemented null, bang-bang, PID and sampling-MPC baselines; tuned PID on a separate tuning split, selecting 39 C, Kp=0.6 and Ki=Kd=0.
- Confirmed the no-cooling hardness gate and fixed the eight evaluation episodes for separate supervisor-enabled and supervisor-disabled comparisons.

Challenges/Issues:

- Making cooling necessary in demanding cases while retaining a mild scenario that is safe without cooling.
- Normalizing observations and balancing reward terms without hiding unsafe raw-policy behaviour behind the supervisor.
- Accounting for MPC's access to the nominal simulator state and future demand when interpreting comparisons.

Plan for Next Week: Train PPO, SAC and TD3 across five seeds, evaluate every controller on the fixed manifest, and retain checkpoints, logs and per-episode metrics.

Guide's Remarks:

Signature of Student
Harshit Singh Parmar

Signature of Guide
[Dr. Pranav Srivastava]

Retrospective milestone grouping; reporting dates and weekly allocation require confirmation.

IILM UNIVERSITY, GREATER NOIDA
SCHOOL OF COMPUTER SCIENCE & ENGINEERING
WEEKLY PROGRESS REPORT
Even semester 2026-2027

B.Tech. Major Project Details

Project Id	**26P_3CSE389**
Project Title	**Reinforcement Learning-Based Dynamic Battery Thermal Management for Software Defined Electric Vehicles**
Reporting period	_____
Week No.	**4**

Work Completed This Week:

- Completed five saved training runs each for PPO, SAC and TD3: SAC/TD3 reached 20,000 steps per seed and PPO reached its 20,480-step rollout boundary.
- Evaluated saved policies and baselines on the same eight episodes, recording costs, cooling energy, temperature violations and supervisor interventions separately.
- Documented the measured limitation: the present learned policies do not beat tuned PID on mean supervised cost and violate the band when supervision is disabled.
- Verified fifteen Python tests and reproduced 288 saved non-MPC evaluation outcomes; a separate MPC HWFET spot-check also matched its recorded result.
- Completed the recorded laptop-CPU policy-export benchmark and built a working dashboard for fresh Python simulations, comparisons and inspectable evidence downloads.
- Prepared the corrected research manuscript, figures and fifteen-reference bibliography, including four verified 2025-2026 studies; checked its nine-page PDF and synchronized source.

Challenges/Issues:

- Improving intrinsic learned-policy safety and cost performance beyond the current short training budget and tuned PID baseline.
- Distinguishing simulated outcomes and laptop inference timings from physical battery validation or automotive-ECU certification.
- Finalizing venue requirements and evaluator-accessible repository/data/demo links, which remain undecided.

Plan for Next Week: Investigate longer training and robustness checks, refine controller/reward settings, finalize sharing and submission details, and prepare the live simulation demonstration.

Guide's Remarks:

Signature of Student
Harshit Singh Parmar

Signature of Guide
[Dr. Pranav Srivastava]

Retrospective milestone grouping; reporting dates and weekly allocation require confirmation.